# DSA Foundation Practice Sheet

Focus Areas: Loops, Nested Loops, Arrays, Strings, Character Indexing, Frequency Counting, Dry Running Logic

## SECTION 1 — BASIC LOOPS

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|---|
| 1. Print 1 to n |
| 2. Print n to 1 |
| 3. Print even numbers |
| 4. Print odd numbers |
| 5. Sum of first n numbers |
| 6. Sum of even numbers |
| 7. Sum of odd numbers |
| 8. Multiplication table |
| 9. Count digits |
| 10. Reverse number |
| 11. Palindrome number |
| 12. Armstrong number |
| 13. Strong number |
| 14. Factorial |
| 15. Fibonacci series |
| 16. Prime number check |
| 17. Perfect number |
| 18. GCD of two numbers |
| 19. LCM of two numbers |
| 20. Power calculation |

## SECTION 2 — NESTED LOOPS + PATTERNS

| |
|---|
| 1. Square pattern |
| 2. Hollow square |
| 3. Right triangle |
| 4. Inverted triangle |
| 5. Pyramid |
| 6. Inverted pyramid |
| 7. Diamond pattern |
| 8. Butterfly pattern |
| 9. Sandglass pattern |
| 10. Hollow pyramid |
| 11. Floyd’s triangle |
| 12. Pascal triangle |
| 13. Number pyramid |
| 14. Palindrome pyramid |
| 15. Zig-zag pattern |
| 16. Alphabet triangle |
| 17. Reverse alphabet triangle |
| 18. Character pyramid |
| 19. Continuous character pattern |
| 20. Hollow character square |

## SECTION 3 — ARRAY FUNDAMENTALS

| 1. Print array elements |
| --- |
| 2. Sum of array |
| 3. Average of array |
| 4. Maximum element |
| 5. Minimum element |
| 6. Second largest |
| 7. Second smallest |
| 8. Count even/odd |
| 9. Reverse array |
| 10. Copy array |
| 11. Linear search |
| 12. Binary search |
| 13. Find index of element |
| 14. Check element exists |
| 15. Frequency of element |
| 16. Check sorted array |
| 17. Move zeros |
| 18. Remove duplicates |
| 19. Left rotate array |
| 20. Right rotate array |
| 21. Merge two arrays |
| 22. Union of arrays |
| 23. Intersection of arrays |
| 24. Missing number |
| 25. Find duplicate number |

## SECTION 4 — STRING FUNDAMENTALS

| 1. Reverse string |
| --- |
| 2. Palindrome string |
| 3. Count vowels |
| 4. Count consonants |
| 5. Count words |
| 6. Remove spaces |
| 7. Convert uppercase to lowercase |
| 8. Convert lowercase to uppercase |
| 9. Toggle case |
| 10. Check substring |
| 11. Print characters one by one |
| 12. Count uppercase letters |
| 13. Count lowercase letters |
| 14. Count digits in string |
| 15. Count special characters |
| 16. Valid palindrome |
| 17. Reverse words in string |
| 18. Longest word |
| 19. First non-repeating character |
| 20. Find duplicate characters |
| 21. Remove duplicate characters |
| 22. String compression |
| 23. Valid anagram |
| 24. Pangram |
| 25. Jewels and Stones |

## SECTION 5 — CHARACTER INDEXING

| 1. Count frequency of lowercase letters |
| --- |
| 2. Count frequency of uppercase letters |
| 3. Count all characters |
| 4. Most frequent character |
| 5. Least frequent character |
| 6. First unique character |
| 7. Find repeated characters |
| 8. Check two strings are anagram |
| 9. Ransom Note |
| 10. Find the Difference |

## SECTION 6 — HASHSET / HASHMAP THINKING

| 1. Contains Duplicate |
| --- |
| 2. Happy Number |
| 3. Intersection of Arrays |
| 4. Longest Consecutive Sequence |
| 5. Unique email addresses |
| 6. Two Sum |
| 7. Majority Element |
| 8. Frequency count |
| 9. Group Anagrams |
| 10. Word Pattern |

## SECTION 7 — TWO POINTERS

| 1. Reverse String |
| --- |
| 2. Valid Palindrome |
| 3. Move Zeroes |
| 4. Remove duplicates sorted array |
| 5. Merge sorted arrays |
| 6. Container with most water |
| 7. Two sum sorted array |
| 8. Squares of sorted array |
| 9. Backspace string compare |
| 10. 3Sum (later) |

## SECTION 8 — SLIDING WINDOW

| 1. Maximum sum subarray |
| --- |
| 2. Average of subarray |
| 3. Longest substring without repeating |
| 4. Maximum vowels in substring |
| 5. Permutation in string |
| 6. Minimum window substring (later) |
| 7. Fruits into baskets |
| 8. Longest repeating replacement |
| 9. Binary subarrays |
| 10. Fixed window sum |

## SECTION 9 — DRY RUN TRAINING

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|---|
| 1. Write input |
| 2. Track variable values after each iteration |
| 3. Track loop movement |
| 4. Predict final output |
| 5. Explain why answer works |